



# Macroeconomic Data: Trends, Cycles, and the stylized facts of the Business Cycle

Data are hyperlinked to original sources

Reference: Chapters 1-3, Williamson

- Skim Chapters 1-2

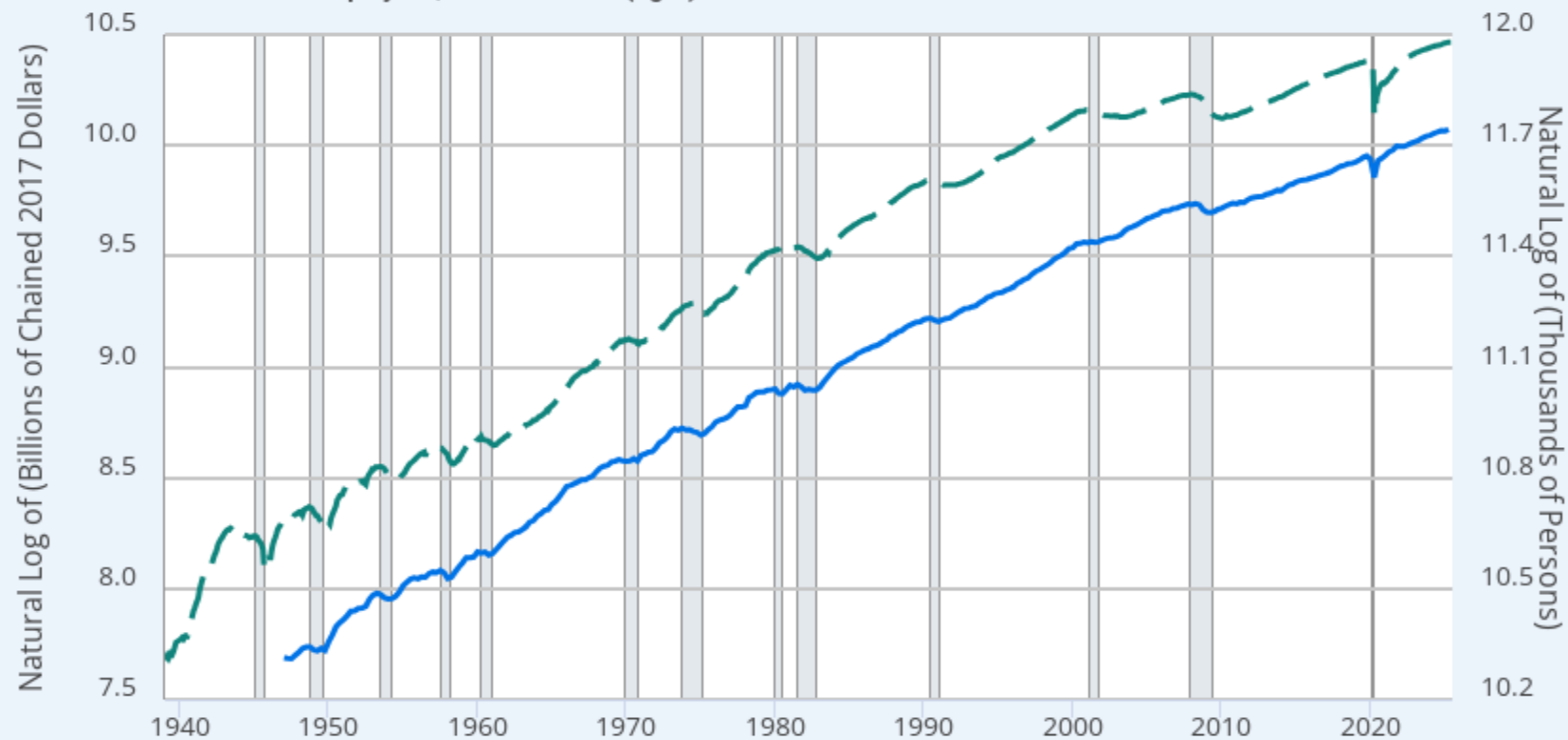
Things you can skip from Chapter 3:

- Edition 6: The sections on “The Price Level and Inflation” and “Seasonal Adjustment”



**FRED** 

— Real Gross Domestic Product (left)  
— All Employees, Total Nonfarm (right)



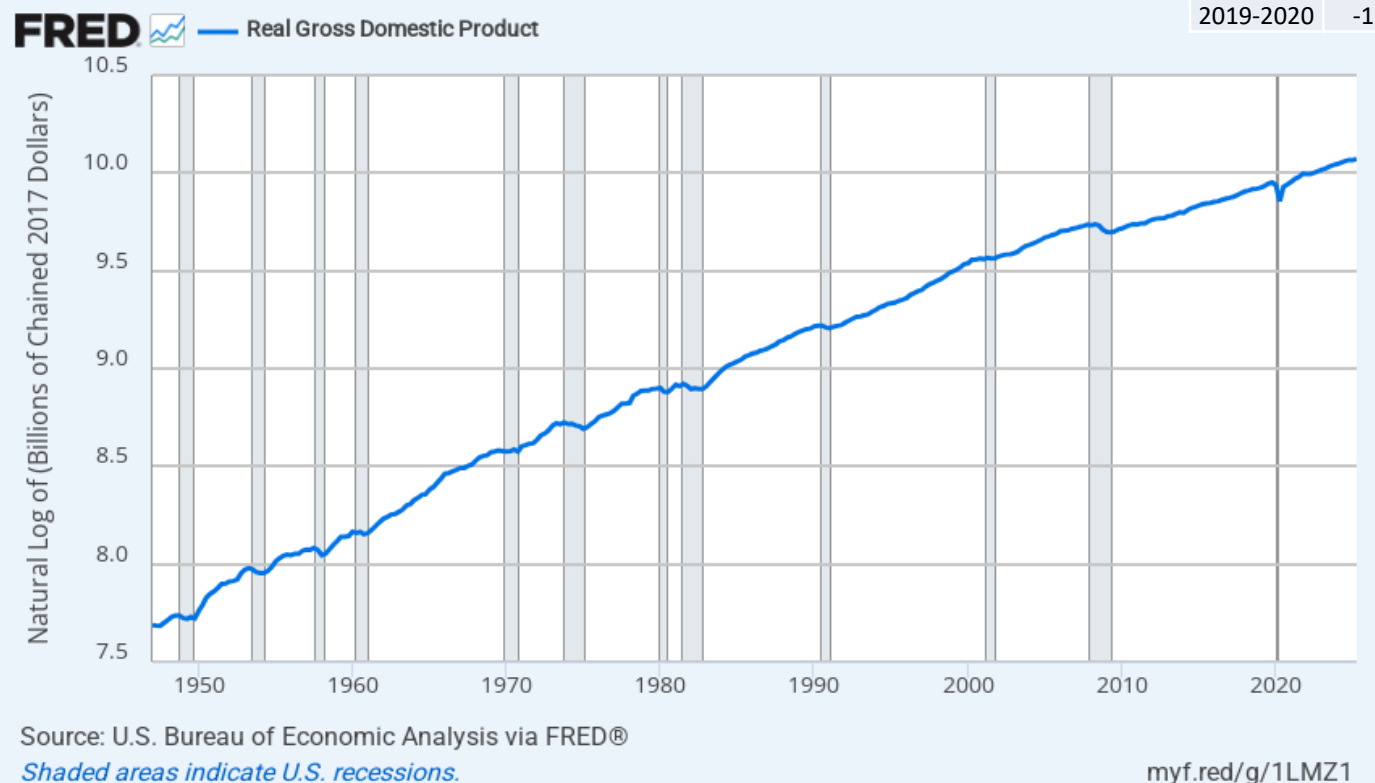
Sources: U.S. Bureau of Economic Analysis; U.S. Bureau of Labor Statistics via FRED®

*Shaded areas indicate U.S. recessions.*

[myf.red/g/1LMYr](https://myf.red/g/1LMYr)

**Trend growth was higher in the post-war 20<sup>th</sup> Century than it has been recently, even excluding the Pandemic**

|           |        |
|-----------|--------|
| 1951-1970 | 3.80%  |
| 1971-1990 | 3.30%  |
| 1991-2000 | 3.60%  |
| 2001-2010 | 1.80%  |
| 2011-2020 | 1.14%  |
| 2021-2025 | 2.50%  |
|           |        |
| 2011-2019 | 2.20%  |
| 2019-2020 | -1.50% |

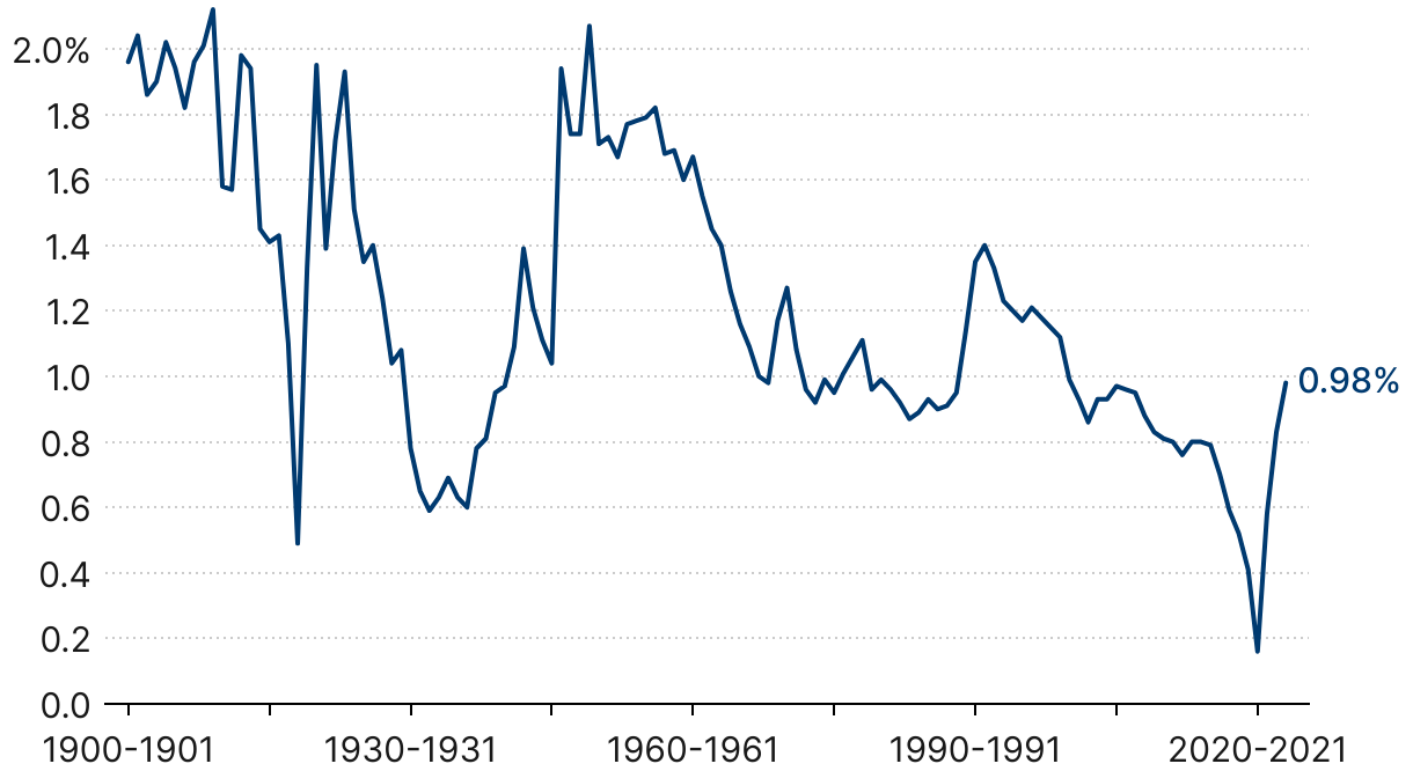


**Recessions were more frequent in the post-war 20<sup>th</sup> Century than they have been recently**



## US annual population growth

Years 1900-01 to 2023-24\*



**Source:** William H. Frey analysis of Census Bureau historical population estimates including 2020-24 annual estimates, released December 19, 2024

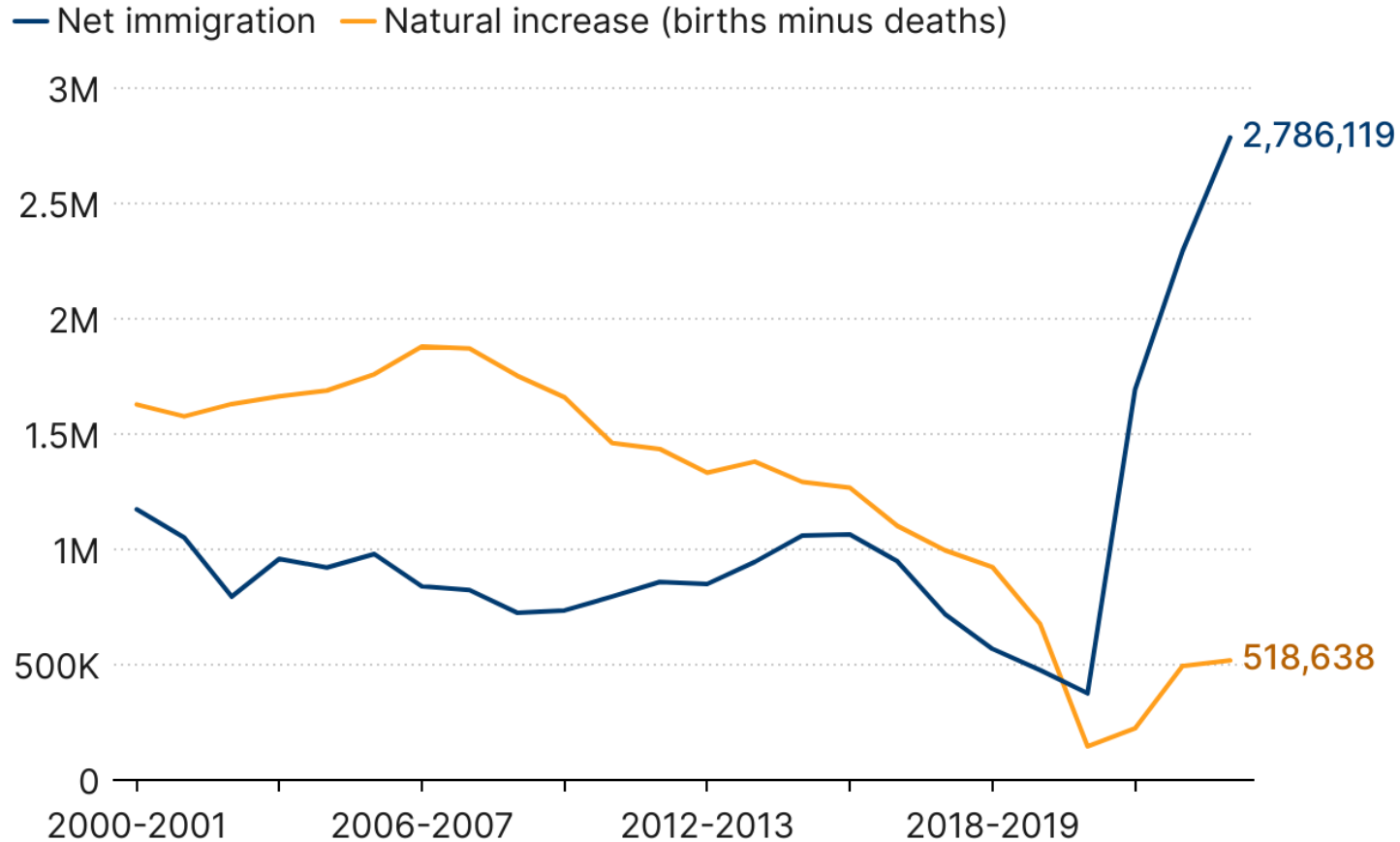
**Note:** \*July 1 to June 30 of each year

**B** | Brookings Metro



## US net immigration and natural increase

2000-01 through 2023-24



**Source:** William H. Frey analysis of Census Bureau population estimates: vintage 2020 and vintage 2024, released December 19, 2024

**Note:** \*July 1 to June 30 of each year

**B** | Brookings Metro

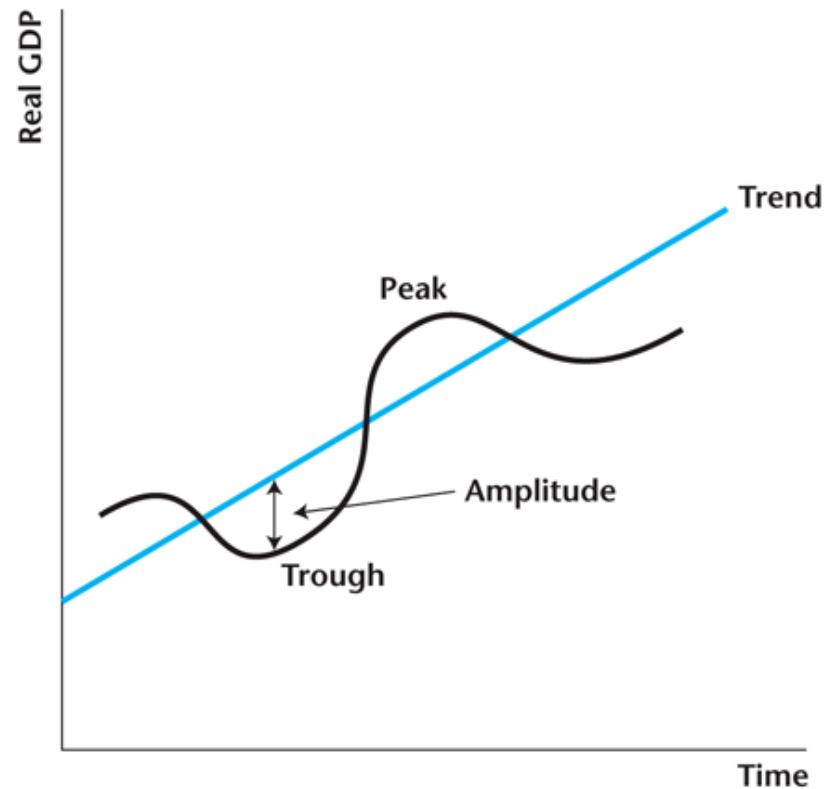


# Business cycles



## Figure 3.1

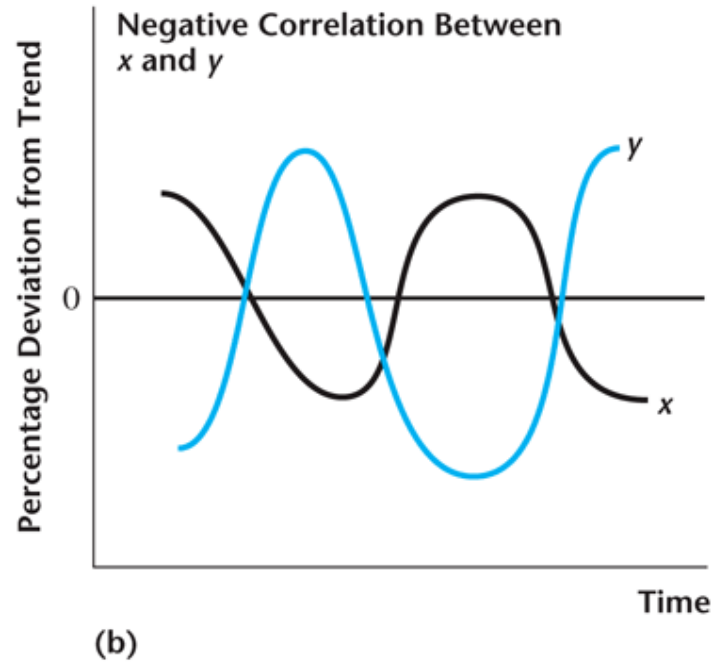
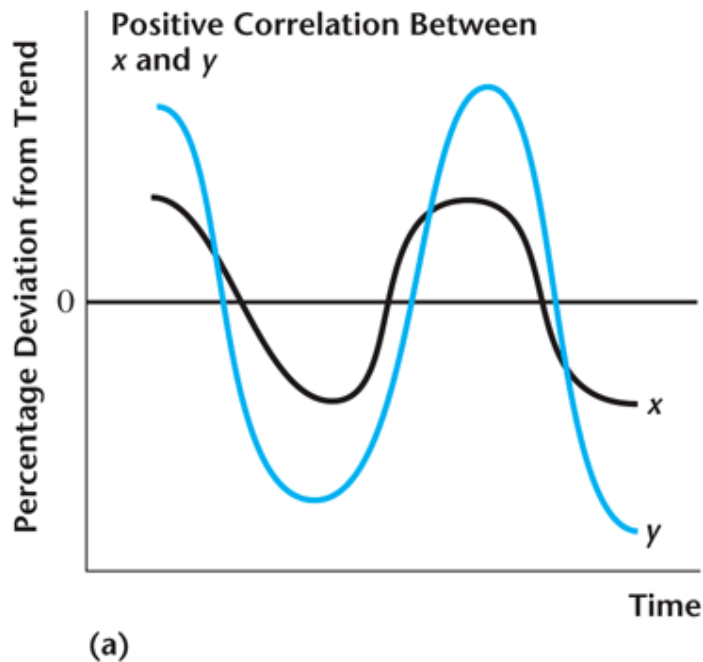
### Idealized Business Cycles





## Figure 3.3

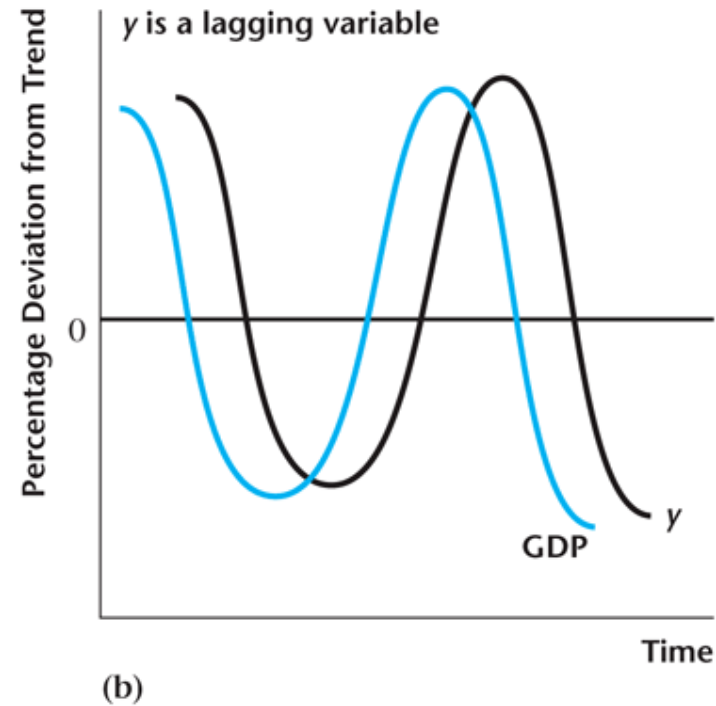
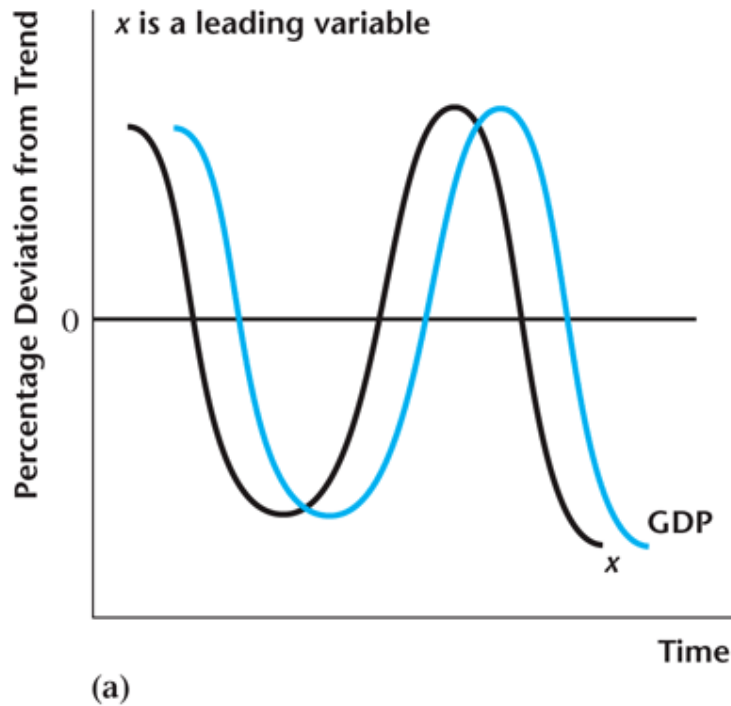
### Time Series Plots of $x$ and $y$



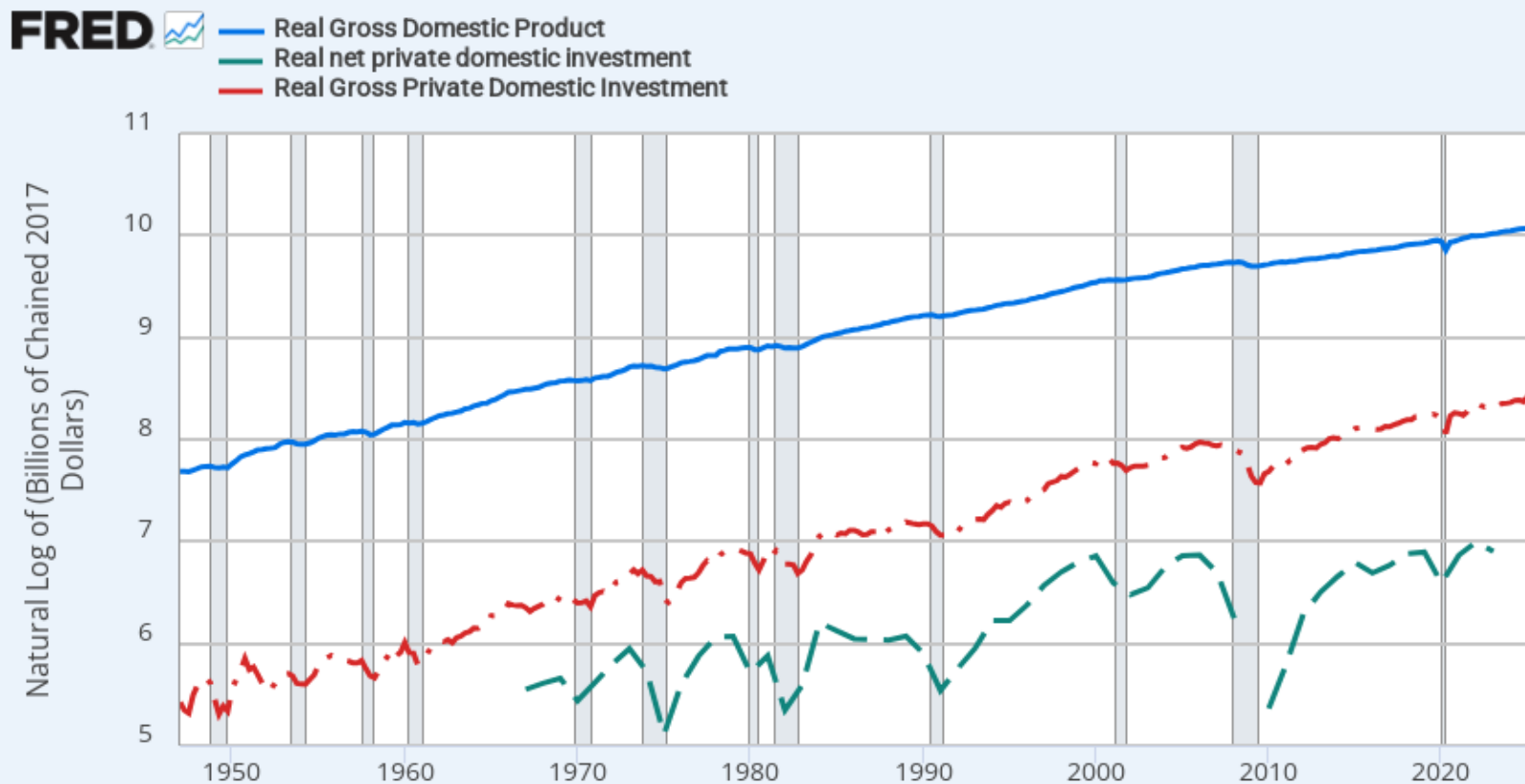


## Figure 3.7

### Leading and Lagging Variables



# Investment is relatively volatile

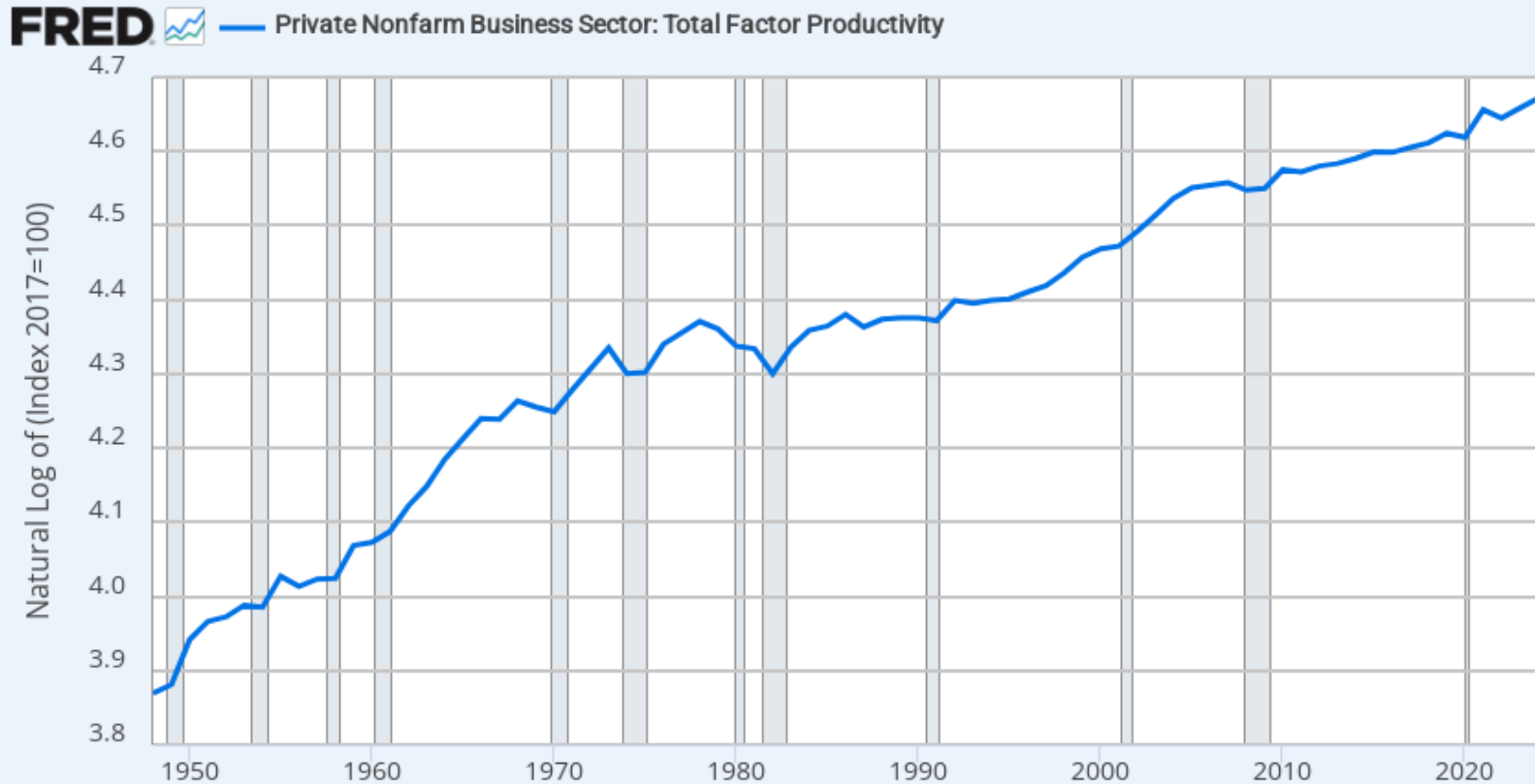


Source: U.S. Bureau of Economic Analysis via FRED®

*Shaded areas indicate U.S. recessions.*

[myf.red/g/1LN6u](https://myf.red/g/1LN6u)

# Total Factor Productivity is pro-cyclical



Source: U.S. Bureau of Labor Statistics via FRED®

*Shaded areas indicate U.S. recessions.*

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# Table 3.1

## Correlation Coefficients and Variability of Percentage Deviations from Trend

|                            | Correlation Coefficient | Standard Deviation (% of S.D. of GDP) |
|----------------------------|-------------------------|---------------------------------------|
| Consumption                | 0.77                    | 77                                    |
| Investment                 | 0.80                    | 301                                   |
| Employment                 | 0.78                    | 65                                    |
| Average Labor Productivity | 0.77                    | 63                                    |

## Table 3.2 Summary of Business Cycle Facts

|                            | Cyclicalities | Lead/Lag   | Variation Relative to GDP |
|----------------------------|---------------|------------|---------------------------|
| Consumption                | Procyclical   | Coincident | Smaller                   |
| Investment                 | Procyclical   | Coincident | Larger                    |
| Employment                 | Procyclical   | Lagging    | Smaller                   |
| Real Wage                  | Procyclical   | ?          | ?                         |
| Average Labor Productivity | Procyclical   | Coincident | Smaller                   |